

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile

devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8-or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for

redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the

access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and	Good collaboration	Limited participation	No collaboration or contribution

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		contribution (if team-based)			

Q1. Star Topology – 20-Person Startup

Answer:

A **star topology** is the most suitable choice for a 20-person startup because every device is connected to a central managed switch. A **24-port Layer 2/Layer 3 managed switch** can connect computers, printers and VoIP phones while providing ports for future expansion. Each workstation should be connected using **Cat6 UTP cable**, supporting Gigabit Ethernet up to 100 m. The central switch connects to a **router/firewall**, which provides Internet connectivity, DHCP and NAT. A wireless access point can also be connected to the switch for laptops and mobile devices.

The major advantage is **fault isolation**. If one workstation cable fails, only that device is affected and the remaining employees continue working. Centralized cabling also makes troubleshooting and future expansion easier.

Implementation: Configure the switch with appropriate ports, connect all 20 devices using Cat6 cables, configure the router for DHCP/NAT, and configure the wireless access point.

Testing: Test connectivity between workstations, Internet access and failure of an individual cable. The expected result is that failure of one workstation does not affect the remaining network.

Q2. Mesh Topology – Banking Network

Answer:

For a bank requiring **near-zero downtime**, a **full or partial mesh topology** is appropriate. In a full mesh, every major network device has a direct connection to other devices, providing multiple paths for communication.

A practical banking implementation uses a **partial mesh at the core layer**, connecting core switches, routers and firewalls redundantly. Access and distribution switches should have redundant uplinks to at least two core switches.

Dual enterprise routers can provide redundant Internet connections using **BGP**. Fiber-optic cables should be used between core devices because they provide high bandwidth, low latency and immunity to electromagnetic interference. Single-mode fiber is suitable for longer building-to-building connections, while OM4 multimode fiber can be used inside a data center.

Implementation: Configure redundant routers, core switches, fiber links and routing protocols such as **OSPF/BGP**.

Testing: Disconnect one network link and verify that traffic automatically uses another available path. This demonstrates fault tolerance and high availability.

Q3. Bus Topology – Small Classroom Lab

Answer:

For a small classroom with a limited budget, a **bus topology** can provide a low-cost networking solution because multiple computers share a common communication path.

A traditional implementation uses a **50-ohm RG-58 coaxial cable**, T-connectors, BNC connectors and 50-ohm terminators at both ends. A modern low-cost alternative can use Cat5e/Cat6 connections with inexpensive networking equipment.

The major advantage is **low hardware cost**, while the main limitation is that a failure in the common cable can affect all connected computers. The topology is therefore suitable for a small, supervised classroom with light network usage such as browsing and file sharing.

Implementation: Connect the classroom systems to the common communication medium and configure IP addresses for communication.

Testing: Verify file sharing and Internet connectivity between systems. Simulate a backbone failure and observe that connected systems lose network connectivity.

Q4. Hybrid Topology – University Campus

Answer:

A large university campus requires a **hybrid topology** because thousands of users are distributed across multiple buildings. A single topology cannot efficiently satisfy requirements for scalability, reliability and cost.

The proposed design combines:

- **Ring/partial mesh** at the campus backbone for redundancy.
- **Star topology** inside individual buildings for easy management.
- **Bus/tree structures** where appropriate inside laboratories.
- **Fiber-optic backbone** between buildings.
- **Cat6 cables** for end-user devices.
- **PoE wireless access points** connected to access switches.

The campus can follow a **three-layer architecture: Core, Distribution and Access**. Core routers and multilayer switches can be connected using high-speed fiber. Each building can have a distribution switch with dual fiber uplinks to the campus core. Floor-level access switches then connect individual computers through Cat6 cables.

Implementation: Configure VLANs, routing, redundant fiber links, access switches and wireless access points.

Testing: Disconnect one backbone link and verify that the building continues to communicate through the redundant path. New buildings should also be tested for integration without disturbing existing networks.

Q5. Multi-Department Company Network

Answer:

A company with departments such as **HR, Finance, Sales and IT** should use a hierarchical network architecture with **VLANs, switches and routers**.

Each department is connected to a dedicated **Layer 2 access switch**. Devices within each department communicate through their respective VLAN. These access switches connect to a **Layer 3 core switch or router**, which performs **inter-VLAN routing** and allows controlled communication between departments.

For example:

Department VLAN

HR	VLAN 10
Finance	VLAN 20
Sales	VLAN 30
IT	VLAN 40

The router/firewall provides Internet connectivity and security. VLAN separation reduces unnecessary broadcast traffic and improves security. **Subnetting, redundant links and backup paths** can further improve reliability and scalability.

Implementation: Create department VLANs, assign switch ports, configure trunk links, configure inter-VLAN routing and apply appropriate IP addressing.

Testing: Verify that devices within the same VLAN communicate correctly, test inter-VLAN communication through the Layer 3 device, and verify that unauthorized departmental access is restricted.